

HD189733b

R-0060

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HD189733_230704 · created 2026-07-15T14:38:05+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460129.8214 +/- 0.0017 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.155 +/- 0.016
Transit depth (Rp/Rs)^2	2.41 +/- 0.48 [%]
Orbital Inclination (inc)	85.31 +/- 0.92
Airmass coefficient 1 (a1)	1.0214 +/- 0.0098
Airmass coefficient 2 (a2)	-0.021 +/- 0.025
Scatter in the residuals of the lightcurve fit is	0.97 %
Best Comparison Star	#1 - [262, 106]
Optimal Aperture	6.81
Optimal Annulus	11.9
Transit Duration (day)	0.07 +/- 0.01

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.155 ± 0.016 vs 0.15534 (deviation 0.02σ)
Duration fit vs published	0.07 d vs 0.0754 d (deviation 0.54σ)
Mid-time O–C	2.6 min
Depth SNR	9.7
Residual scatter	0.97 %

Light curve

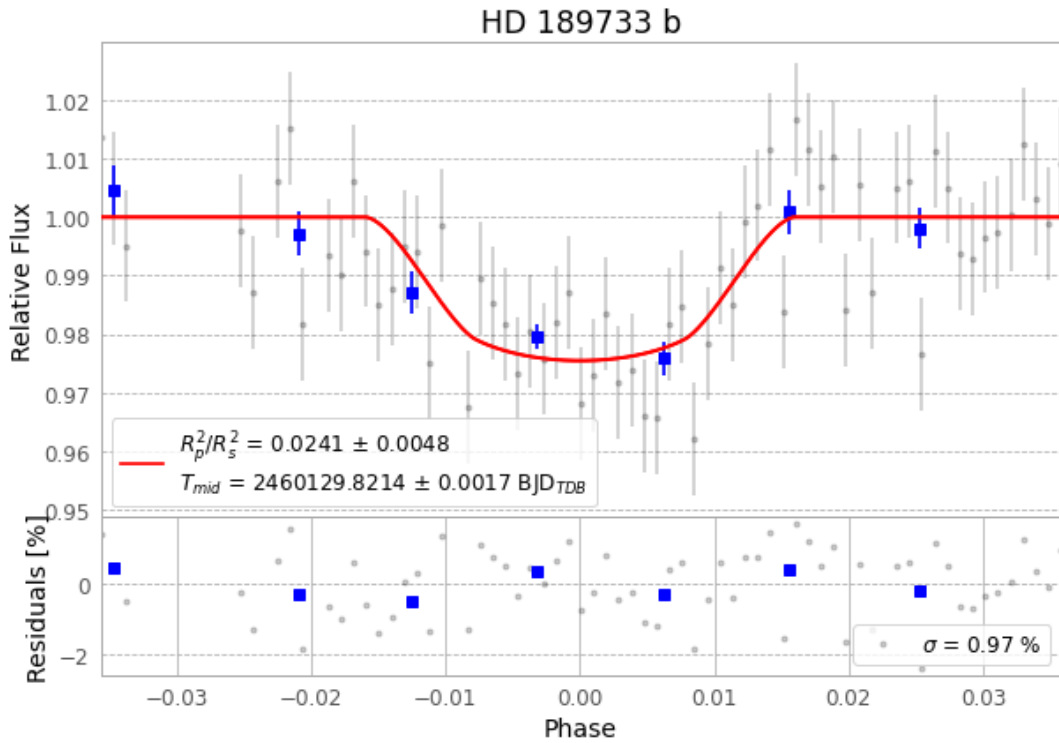


Figure 1 — FinalLightCurve_HD 189733 b_2023-07-03.png (SHA-256 7c5233a27ba8504e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [298, 201], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 eac3cf478dd11c87...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

72 FITS frames (47 MB), HD189733230704054215.FITS → HD189733230704093015.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HD189733-230704.log (6b03c7df3ae0...), FinalLightCurve_HD 189733 b_2023-07-03.png (7c5233a27ba8...), FinalLightCurve_HD 189733 b_2023-07-03.csv (b3342d5f5618...)

Compute resources

Wall time 146.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-15 16:40Z.